

AL-AZHARI LIBRARY — PHASE 16 COMPLETION REPORT

1. Executive Summary & Verification Metrics

Phase 16 (Full Testing — Integration and Concurrency Suite — Acceptance Criteria Pass) has been completed for the AL-AZHARI LIBRARY backend.

All quality gates, security regressions, cross-module state machines, race condition matrices, transaction failure injection tests, and database migrations passed with **0 errors, 0 warnings, and 0 regressions**.

Verified Quality Gates

Quality Gate	Command	Result	Metrics
TypeScript Compilation	npx tsc --noEmit	PASS	0 errors
ESLint Quality & Style	npm run lint	PASS	0 warnings, 0 errors
Production Build	npm run build	PASS	Clean build to dist/
Dedicated Phase 16 Suites	npx jest tests/workflows/ tests/integration/ tests/api/ tests/unit/migrations-regression.test.ts	PASS	6 suites, 35 tests passed
Full Repository Test Suite	npm test	PASS	117 suites passed, 117 total 858 tests passed, 858 total (66.88s)

2. Strict Scope Boundary Confirmation

- **Phase 16 ONLY:** All work strictly adhered to Phase 16 boundaries.
- **Phase 17 (Security Hardening): NOT implemented** (deferred to Phase 17 per authoritative plan).
- **Phase 18 (Deployment & Production Configuration): NOT implemented** (deferred to Phase 18).
- **Zero Business Logic Redesign:** Existing domain rules, schemas, and state machines remain preserved.
- **Open Decisions Preserved Unmodified:**
 - **OD-03** (Return Shipping Label Generation): Preserved as manual admin review and triage.
 - **OD-04** (Guest Order Lookup Rate Limiter): Kept established thresholds.
 - **OD-05** (External Payment Webhook Signature Scheme): Preserved manual bank transfer / receipt upload workflow.
 - **OD-06** (Outbox Change Streams vs Polling): Retained atomic lease claiming with recoverable polling workers.
 - **OD-17** (Audit Retention Policy): Retained append-only immutable retention with no destructive automated purges.

3. Dedicated Phase 16 Test Suites

Six dedicated suites (35 tests total) were authored and verified to stress cross-module contracts, concurrency, authorization, horizontal data isolation, transaction rollbacks, and schema migrations:

A. Cross-Module Workflows

File: tests/workflows/cross-module-journeys.test.ts (5 tests — PASS)

- **Journey A (Full Standard Customer Order Lifecycle):** User registration -> address creation -> cart management -> coupon validation -> checkout with idempotency -> manual bank transfer proof upload -> admin payment review & acceptance -> multi-stage fulfillment (processing -> shipped -> delivered) -> inventory deduction -> outbox dispatch.
- **Journey B (Guest Checkout & Order Tracking):** Guest cart -> guest checkout with delivery snapshot -> secure order lookup via reference + guest secret token.
- **Journey C (Student Services Workflow):** Custom research formatting service request -> admin price quotation -> student quotation acceptance -> service order progression -> delivery.
- **Journey D (Returns & Refunds Lifecycle):** Return request submission on delivered order -> admin review & approval -> physical receipt & refund completion -> stock restoration -> audit log entry.
- **Journey E (Outbox Worker & Email Adapter Resilience):** Asynchronous event generation -> outbox worker leasing -> simulated SMTP delivery -> lease completion.

B. High-Concurrency & Race Condition Matrix

File: tests/integration/concurrency-matrix.test.ts (8 tests — PASS)

- **Last-Stock Race Condition:** 2 concurrent checkouts competing for the 1 remaining unit in stock (stock = 1). Exactly 1 succeeds (201); the other fails (400 / 409). Remaining stock ends at exactly 0 , never negative.
- **Checkout Idempotency Deduplication:** 2 concurrent checkouts with identical Idempotency-Key headers. Exactly 1 order is created; duplicate execution is prevented.
- **Cart Price & Stock Revalidation Race:** Product price drifts while in cart; transaction revalidates current catalog price and rejects stale checkout requests.
- **Concurrent Quotation Decision:** Simultaneous accept/reject requests on the same quotation. Exactly 1 state transition succeeds.
- **Concurrent Duplicate Refund:** Simultaneous refund completions on the same return request are safely deduplicated.
- **Multi-Worker Outbox Claim Competition:** Multiple concurrent worker instances competing for pending outbox messages via atomic findOneAndUpdate with lease timestamps; 0 duplicate processing.

C. RBAC Authorization Matrix

File: tests/api/rbac-matrix.test.ts (6 tests — PASS)

- **Anonymous Access:** Correctly rejected with 401 Unauthorized on protected endpoints.
- **Customer Isolation:** Customer tokens attempting admin endpoints (POST /admin/products , GET /admin/orders , GET /admin/reports/revenue , GET /admin/audit-logs) receive 403 Forbidden .
- **Admin Granular Gating:** Verified fine-grained permissions (products.write , orders.read , payments.review , services.quote , reports.read , audit.read).
- **Owner Universal Bypass:** Owner role with wildcard (*) successfully accesses all administrative and reporting endpoints.

D. IDOR & Data Isolation Regression

File: tests/api/idor-regression.test.ts (7 tests — PASS)

- **Orders:** Customer B cannot inspect or mutate Customer A's order (403 Forbidden / 404 Not Found).
- **Addresses:** Customer B cannot view, edit, or delete Customer A's saved address.
- **Carts:** Customer B cannot read or clear Customer A's shopping cart.
- **Payment Proofs:** Customer B cannot view Customer A's payment receipts.
- **Service Requests & Quotations:** Customer B cannot read, accept, or reject Customer A's requests/quotations.
- **Return Requests:** Customer B cannot view or tamper with Customer A's returns.
- **Notifications:** Customer B cannot list or mark Customer A's notifications as read.

E. Failure Injection & Rollbacks

File: tests/integration/failure-injection.test.ts (5 tests — PASS)

- **Checkout Mid-Flight Abort:** Fault injected mid-transaction during order item snapshotting. Entire transaction aborts; 0 orders created, cart intact, 0 stock deducted.
- **Order Acceptance Abort:** Fault injected during inventory reservation. Order remains `pending_review`, no stock leaked.
- **Return Approval Abort:** Fault injected during refund record creation. Return status remains pending.
- **Decoupled Realtime Resilience:** Socket.IO emitter failure does not crash or abort the parent database transaction; business operations complete authoritatively.
- **Error Envelope Sanitization:** Malformed requests and internal errors produce uniform `{ success: false, error: { message, code } }` envelopes without leaking MongoDB connection URIs, database credentials, or internal Mongoose stack traces.

F. Migrations Regression & Idempotency

File: tests/unit/migrations-regression.test.ts (4 tests — PASS)

- **Strict Chronological Sequence:** Validates all 13 migrations from `20260927_001` through `20260928_013`.
- **Clean Database Run:** Executes all 13 migrations successfully from scratch on an empty MongoDB database.
- **Repeat-Safe Idempotency:** Second execution detects applied state and executes 0 migrations cleanly.
- **Schema & Index Verification:** Verifies presence of collections and indexes across all 15 functional domains.

4. Defect Log & Invariants Addressed

During the integration and regression suite implementation, the following edge cases were resolved without altering core business rules:

1. Migration Auto-Registration:

- *Issue:* Early migration scripts (`001` through `008`) exported migration objects but did not invoke `registerMigration(...)`, causing `MigrationRunner.getRegisteredMigrations()` to omit them when imported dynamically.
- *Resolution:* Added `registerMigration(...)` calls to scripts `001` – `008` in `src/database/migrations/scripts/`.

2. Duplicate Index Guarding in Migrations:

- *Issue:* `auditLogs` and `shippingRules` had index keys created both via Mongoose schema initialization and migration scripts with differing names (`idx_audit_logs_entity` vs `idx_audit_logs_entity_created`), causing MongoDB error code 85 during sequential test runs.
- *Resolution:* Wrapped index creation calls in defensive `try/catch` blocks in migration scripts, ensuring seamless idempotency whether running on fresh databases or pre-indexed collections.

5. Artifacts and Documentation

- `docs/PHASE-16-FULL-TESTING.md` : Architectural test documentation detailing test layering, environment isolation, failure injection models, and open decisions status.
- `tests/workflows/cross-module-journeys.test.ts`
- `tests/integration/concurrency-matrix.test.ts`
- `tests/api/rbac-matrix.test.ts`
- `tests/api/idor-regression.test.ts`
- `tests/integration/failure-injection.test.ts`
- `tests/unit/migrations-regression.test.ts`

Phase 16 is complete. The repository is ready for Phase 17 (Security Hardening).